

B8 Work, Energy and Power

Strength of Earth's gravity at ground level = 9.81 N kg^{-1} .

- B8.1 A block of mass 1.2 kg slips all the way down a 1.4 m shelf inclined at 30° to the horizontal.
- a) How much GPE does the object lose?
 - b) If the average frictional force is 1.5 N , work out how much work the object does against friction.
- B8.2 A 90 kg cycle and rider travels at a steady speed, covering 35 m in 5 seconds. Frictional forces are constant and are 60 N in total. Work out the power output of the cyclist, assuming 100% efficiency.
- B8.3 A 900 kg roller coaster car begins its journey at ground level with a speed of 35 m s^{-1} . By the time it has travelled 55 m directly upwards, it has a velocity of 11 m s^{-1} . Assuming that the force resisting motion has been constant, how large is it?

CHAPTER B. MECHANICS

Answers

B8.1

a) $1.2 \times 9.81 \times 1.4 \times \sin 30^\circ = 8.24 \text{ J}$

b) $1.5 \times 1.4 = 2.1 \text{ J}$

B8.2 $60 \times \frac{35}{5} = 420 \text{ W}$

B8.3 $\frac{0.5 \times 900 \times (35^2 - 11^2) - 900 \times 9.81 \times 55}{55} = 203.72 \text{ N}$